



Data Vortex

Comprehensive Master Report

Team: Event Horizon

Contents: Exploratory Data Analysis, Data Quality Audit, & Statistical Methodology

📄 Executive EDA Summary & Insights # Exploratory Data Analysis Report — Social Engine / Data Vortex ## Executive Summary This report analyzes the recovered social media dataset comprising **1,501 users** and **12,000 posts** across 5 major social platforms. ### Key Metrics Overview - **Total Recovered Users**: 1,501 - **Total Cleaned Posts**: 12,000 - **Active User Rate**: 99.9% of users posted at least once - **Average Posts per User**: 8.00 (Median: 8) - **Mean Follower Count**: 24,963.9 (Median: 24,741.5) - **Correlation (Followers vs Engagement)**: -0.0183 --- ## 1. User Demographics & Geographic Insights - **Top Locations**: Users originate from globally distributed metropolitan hubs including London, Tokyo, New York, São Paulo, and Paris. - **Language Split**: Balanced multi-lingual presence across English (en), Spanish (es), French (fr), German (de), Japanese (ja), Arabic (ar), and Mandarin (zh). - **Follower Distribution**: Uniform distribution indicating synthetic benchmark generation across tier brackets from 1,000 to 50,000 followers. ![Follower Distribution] (reports/figures/01_follower_distribution.png) ![Top Locations] (reports/figures/02_top_locations.png) --- ## 2. Platform Dynamics & Engagement - **Platform Share**: Posts are evenly distributed across YouTube, Instagram, Twitter, Reddit, and Facebook (~2,400 posts each). - **Engagement Drivers**: - **YouTube & Instagram** drive highest average

-----| | ****Timestamps**** | Mixed Unix Epoch (`1704...`), `DD-MM-YYYY`, and ISO-8601 | 100% Standard ISO-8601 (`YYYY-MM-DDTHH:MM:SS`) | | ****Text Markup**** | Scraped HTML (`

`, ``), HTML entities (`&`), mojibake (`Ã©`) | Pure text sanitized strings | | ****Numeric Types**** | Inconsistent string numbers, placeholders (`NULL`, `?`) | Clean 64-bit integer & float columns | | ****Categoricals**** | Inconsistent casing and sentinel tokens | Cleaned canonical platform names | --- ## 4. Profiling Artifacts - ****Raw Profiling Report****: `reports/raw\_profile.html` (Open locally in a web browser) - ****Cleaned Profiling Report****: `reports/cleaned\_profile.html` (Open locally in a web browser) - ****Comparison Profiling Report****: `reports/comparison\_profile.html` (Open locally in a web browser) - ****Granular Change Log****: [docs/change_log.csv](change_log.csv)

📄 Statistical Decisions & Imputation Rationale # Decisions Log — Social Engine / Data Vortex Hackathon > ****Purpose****: Record all data cleaning decisions, imputation strategies, dropped records, and assumptions with mathematical and statistical justification. --- ## 1. Corruption Patterns Found in Raw Data | Dataset | Column | Pattern Detected | Detection Method | Severity & Impact | |-----|-----|-----|-----|-----|-----| | Users | `account\_created` | Unix epoch timestamps mixed with ISO-8601 strings | Regex & numeric casting inspection | High — parsed to standard `YYYY-MM-DD` | | Users | `follower\_count` | Negative values / corrupted strings / missing entries | Numeric bounds check | Medium — median imputed, negatives clipped | | Users | `location` | Trailing spaces, mixed casing | String inspection | Low — stripped & standardized | | Posts | `platform` | Placeholder strings (`NULL`, `None`, `?`), 1,219 missing | Categorical frequency check | High — filtered to 5 valid platforms | | Posts | `text\_content` | HTML tags (``, `

`), HTML entities (`&`, ``), mojibake (`Ã©`) | Regex regex pattern scan | High — cleaned via HTML/entity stripping | | Posts | `timestamp` | Tri-modal formats: ISO-8601, `DD-MM-YYYY`, and Unix epoch seconds | Composite date parser | High — converted to uniform `YYYY-MM-DDTHH:MM:SS` | | Posts | `likes` / `shares` / `comments` | Numeric string placeholders and 1,858 missing records | Type coercion & null check | Medium — median imputation applied | | Posts | `post\_id` | 360 duplicate post entries | `post\_id` frequency check | Medium — deduplicated keeping first instance | --- ## 2. Cleaning & Imputation Decisions ### Placeholder Replacement - ****Decision****: Replace all sentinel strings (`N/A`, `none`, `null`, `NULL`, `?`, ``, `

``unknown``) with proper ``np.nan``. - ****Justification****: These strings are data-entry artefacts. Coercing to ``np.nan`` ensures correct type inference in downstream statistical operations. Validated via regex exact-string matching (e.g., ``^\s*-\s*$``) to ensure in-text hyphens in legitimate text were preserved. **### Numeric Columns — Median Imputation** - ****Decision****: Impute missing values in numeric columns (``follower_count``, ``likes``, ``shares``, ``comments``) with the column ****median****. - ****Justification****: Engagement metrics are heavily skewed with fat-tail distributions. The median is resistant to outliers (e.g. viral posts), preventing bias propagation (Little & Rubin, 2002). **### Categorical Columns — Explicit Null Preservation & Flagging** - ****Decision****: For ``platform`` and ``text_content``, preserve ``NaN`` / flag invalid categories with an indicator column rather than imputing arbitrary values. - ****Justification****: Imputing text content or platform names where no clear dominant category exists (>30% mode threshold) would introduce phantom signal and synthetic bias. **### Multi-Format Timestamp Parsing** - ****Decision****: Implement a 3-tier date parser: (1) Unix epoch seconds, (2) Day-first ``DD-MM-YYYY``, (3) ISO-8601 strings. - ****Justification****: Captures 100% of valid timestamps across heterogeneous logging sources without data loss. --- **## 3. Rows / Columns Dropped | Stage | Dataset | Target | Count Dropped | Reason |** |-----|-----|-----|-----|-----| | **Deduplication | Posts | Duplicate ``post_id`` | 360 rows | Exact duplicates produced during data transmission retry |** | **Standardization | Text | HTML tags / entities | 11,164 cells | Cleaned noisy formatting markup from scraped content |** --- **## 4. Deliverables → Evaluation Criteria Mapping | Deliverable | Evaluation Criterion Addressed |** |-----|-----| | ``src/clean_data.py``, ``src/run_cleaning.py`` | **Data Handling & Preprocessing Logic |** | ``docs/decisions.md``, ``docs/change_log.csv`` | **Data Cleaning Accuracy & Auditability |** | ``reports/raw_profile.html``, ``reports/cleaned_profile.html`` | **Data Consistency & Standardisation |** | ``docs/profiling_comparison.md`` | **Data Consistency & Standardisation |** | ``notebooks/eda.ipynb``, ``reports/eda_summary.md`` | **EDA Depth & Insight Discovery |** | ``reports/figures/``, markdown commentary | **Insight Interpretation & Clarity |** | ``README.md``, docstrings, modular pipeline | **Code Quality & Documentation |**